



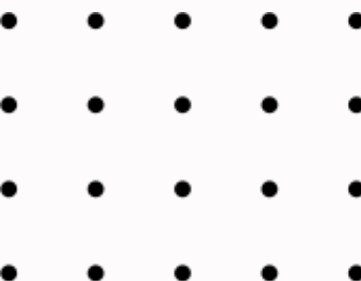
HELP International: Aid Prioritization Using Unsupervised Learning

Comparing PCA vs Autoencoders for Dimensionality Reduction and
Applying Hierarchical Clustering

By **Eliya Christopher Nandi** [0829396]

eliyachristopher.nandi@community.unipa.it

MSc in Statistics and Data Science
Course: Statistical Machine Learning



Background & Problem Setting (HELP International)

- **HELP International** is a humanitarian NGO working to **combat poverty and provide basic amenities** in underdeveloped countries.
- The organization has raised **\$10 million** and needs to **allocate aid strategically** to the countries with the most urgent need.
- As the data scientist, the goal is to support the CEO by using **data-driven segmentation** of countries based on **socio-economic and health indicators**.
- The core challenge is that need is **multidimensional**: countries differ simultaneously in mortality, life expectancy, income, inflation, GDP per capita, and trade/health spending.
- Therefore, we use **unsupervised learning** (no labels) to:
 - (i) compress information (PCA vs Autoencoder) and (ii) group countries (Hierarchical Clustering) to guide prioritization

Dataset & Variables (and why preprocessing is needed)

Dataset summary

- ❑ **Observations:** 167 countries
- ❑ **Features:** 9 numeric indicators + country name
- ❑ **Indicators used:**
child_mort, exports, health, imports, income, inflation,
life_expec, total_fer, gdpp
- ❑ **Data completeness:** No missing values (Total NAs = 0)

Variables are measured in **very different units and scales** (e.g., gdpp vs inflation), so **standardization** is required before distance-based clustering and dimensionality reduction.

Feature engineering

exports, imports, health were originally expressed as % of GDP.

These features were converted them into absolute values using:

$$\text{exports} = \frac{\text{exports} \times \text{gdpp}}{100} \quad \text{imports} = \frac{\text{imports} \times \text{gdpp}}{100}$$
$$\text{health} = \frac{\text{health} \times \text{gdpp}}{100}$$

This transformation expresses trade and health spending in economic magnitude, not percentages, making comparisons more policy-relevant

Scaling / normalization

Z-score standardization was applied to the features

- mean ≈ 0 for each feature

- sd ≈ 1 for each feature

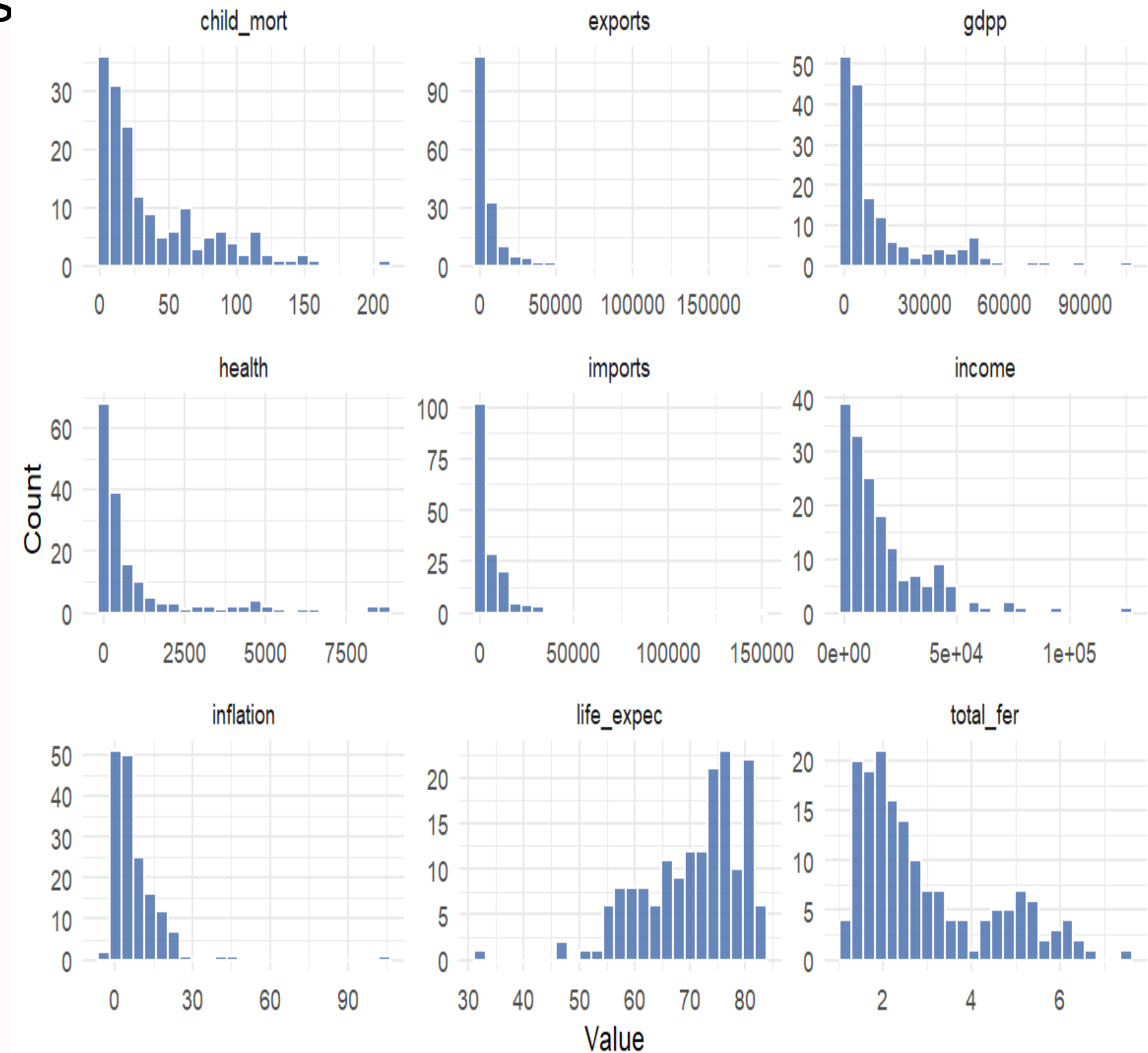
(you showed this with `colMeans = 0` and `sd = 1`)

Note: Standardization prevents variables like **GDP** from dominating Euclidean distances purely due to **larger numeric scale**

Steps:

- Data
- Feature engineering
- Scaling
- Dimension Reduction
- Clustering

Distributions of Features (Raw Scale)

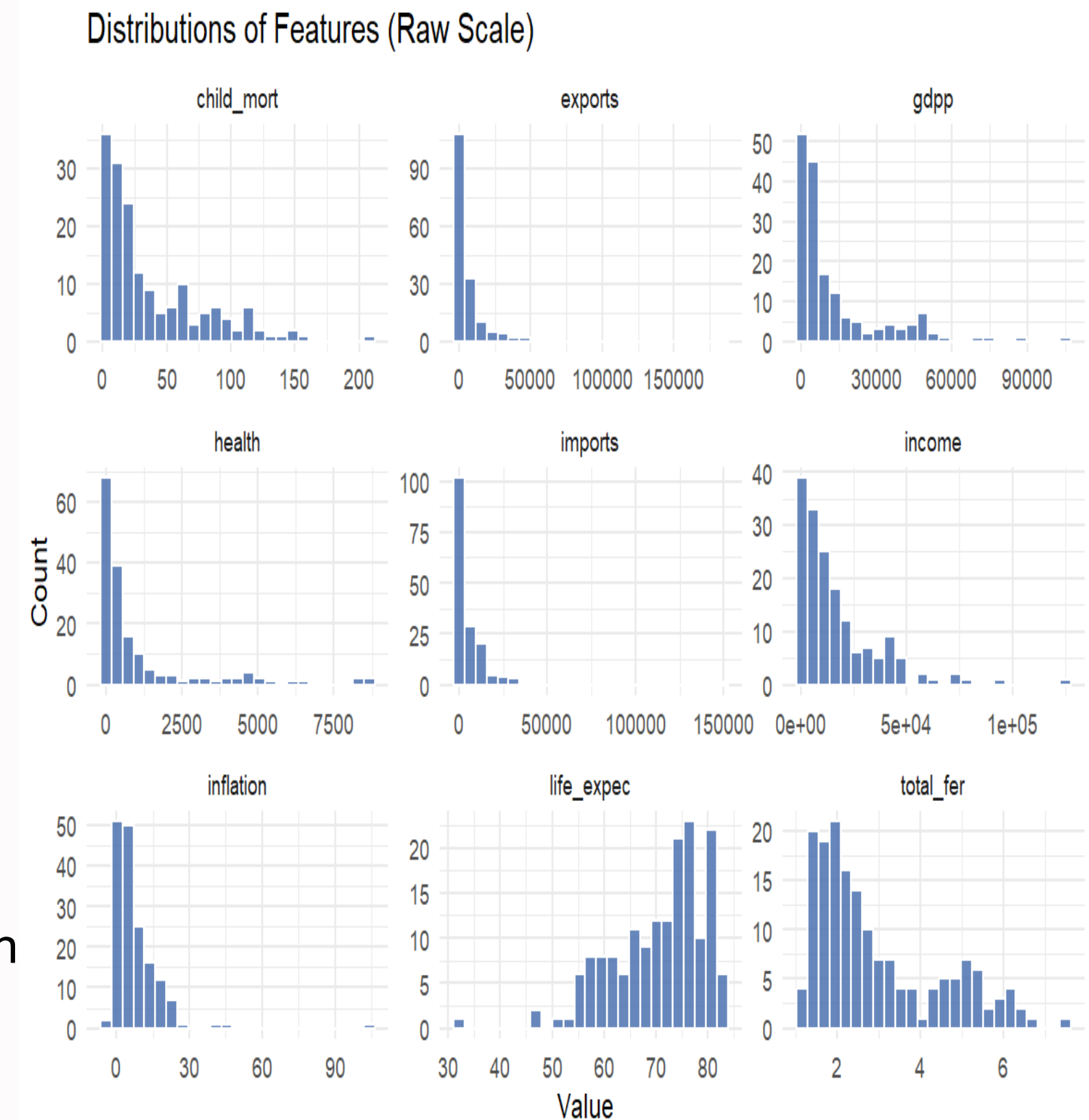


Exploratory Data Analysis (EDA): Distributions & Skewness

EDA: Feature Distributions (Raw Scale) and Skewness Check (Log View)

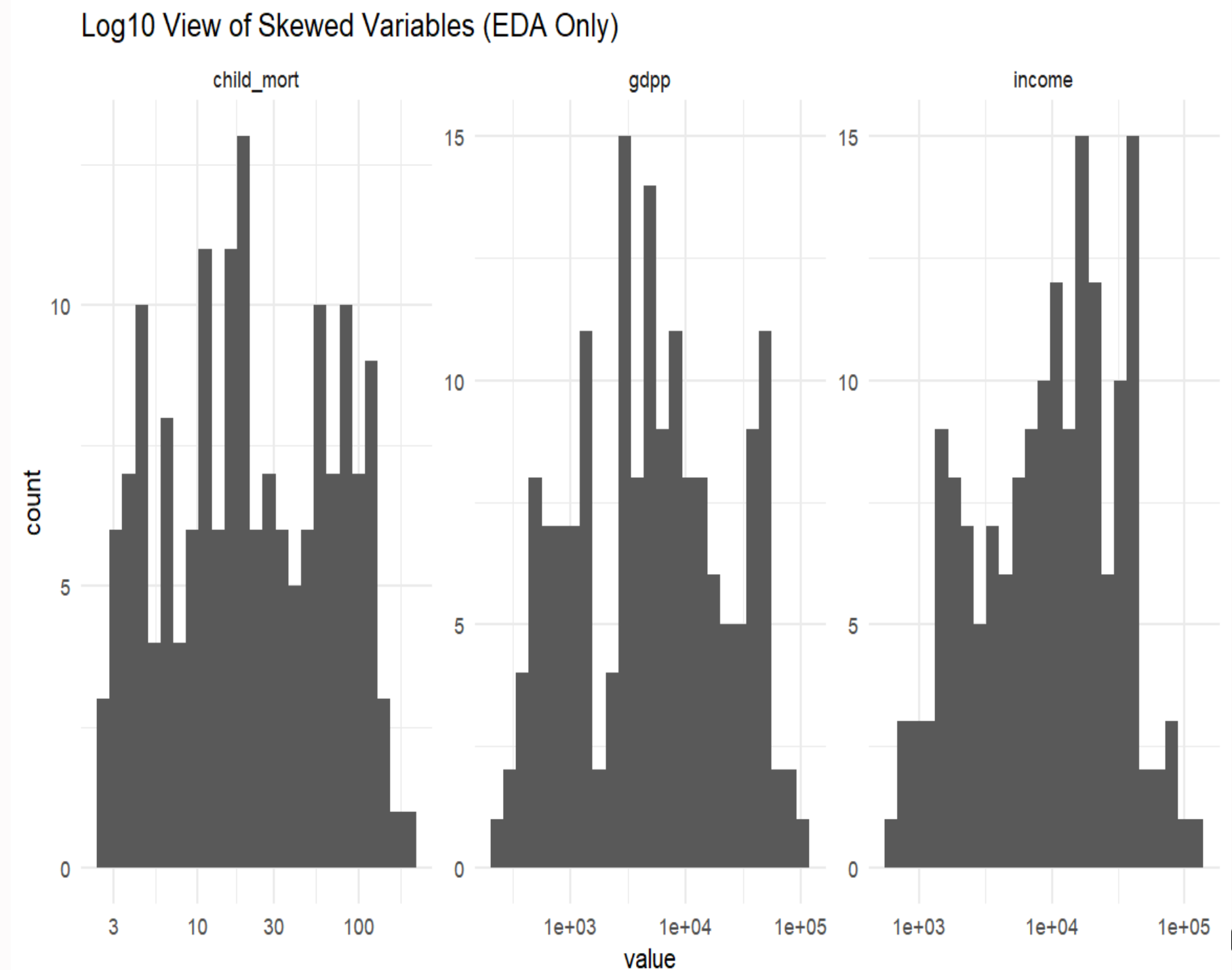
The dataset combines **health outcomes** (child_mort, life_expect, total_fer) and **economic capacity** (gdpp, income, exports, imports, health spending) plus **macro stability** (inflation).

- ❑ Many indicators are **highly right-skewed** (few countries extremely high → long tail).
- ❑ Skewness + scale differences imply:
- ❑ raw values are not directly comparable for distance-based methods,
- ❑ preprocessing choices (scaling + DR) are essential before clustering.
- ❑ A **log10 view** is used only for **EDA** to “zoom in”
- ❑ on low–middle values and see structure that gets compressed on the raw scale.



EDA continuing:

- ❑ A **log10 view** is used only for **EDA** to “zoom in”
- ❑ on low–middle values and see structure that gets compressed on the raw scale.
- ❑ When we view **child_mort**, **gdpp**, **income** on a **log10 scale**, we see the distribution “opens up”:
- ❑ differences among low and middle countries become clearer,
- ❑ the extreme rich countries no longer visually dominate the plot.
- ❑ This confirms the data has **multiplicative / scale-heavy structure** typical for cross-country economics



Note: The data is highly skewed and scale-imbalanced—standardization + dimensionality reduction are essential before clustering

Outlier Detection

Outlier Detection: Univariate (Boxplots) + Multivariate (Z-score Distance)

Why we check outliers in this problem

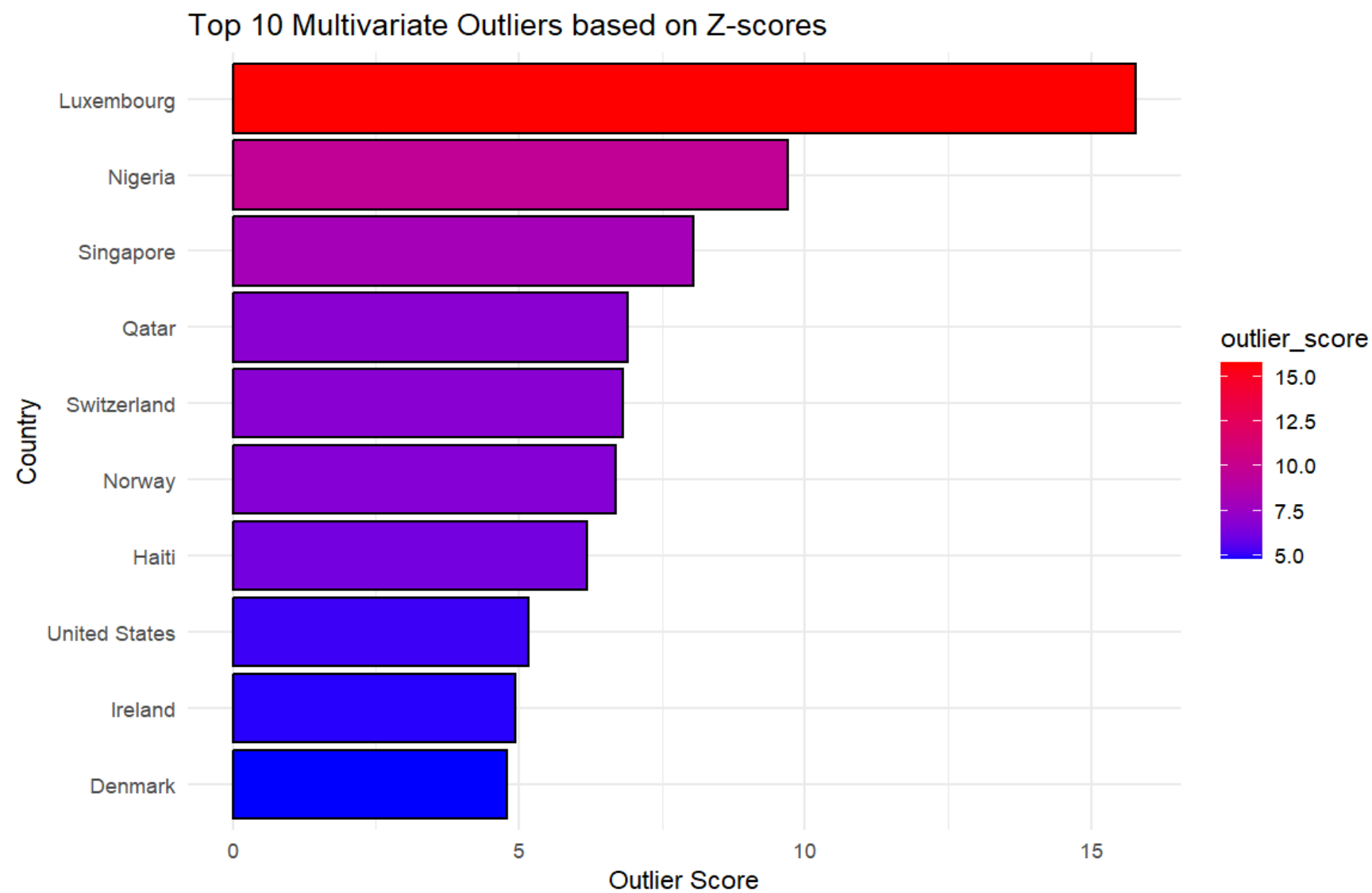
- ❑ In country-level development data, **extreme values are expected** (they often represent real-world heterogeneity, not errors).
- ❑ Outliers strongly affect:
 - ❑ **distance calculations** (hierarchical clustering is distance-based),
 - ❑ **cluster formation** (Ward's method tends to isolate extreme profiles),
- ❑ **dimensionality reduction** (PCA is variance-driven; extremes can influence PC directions).

The implementation involved identifying the following:

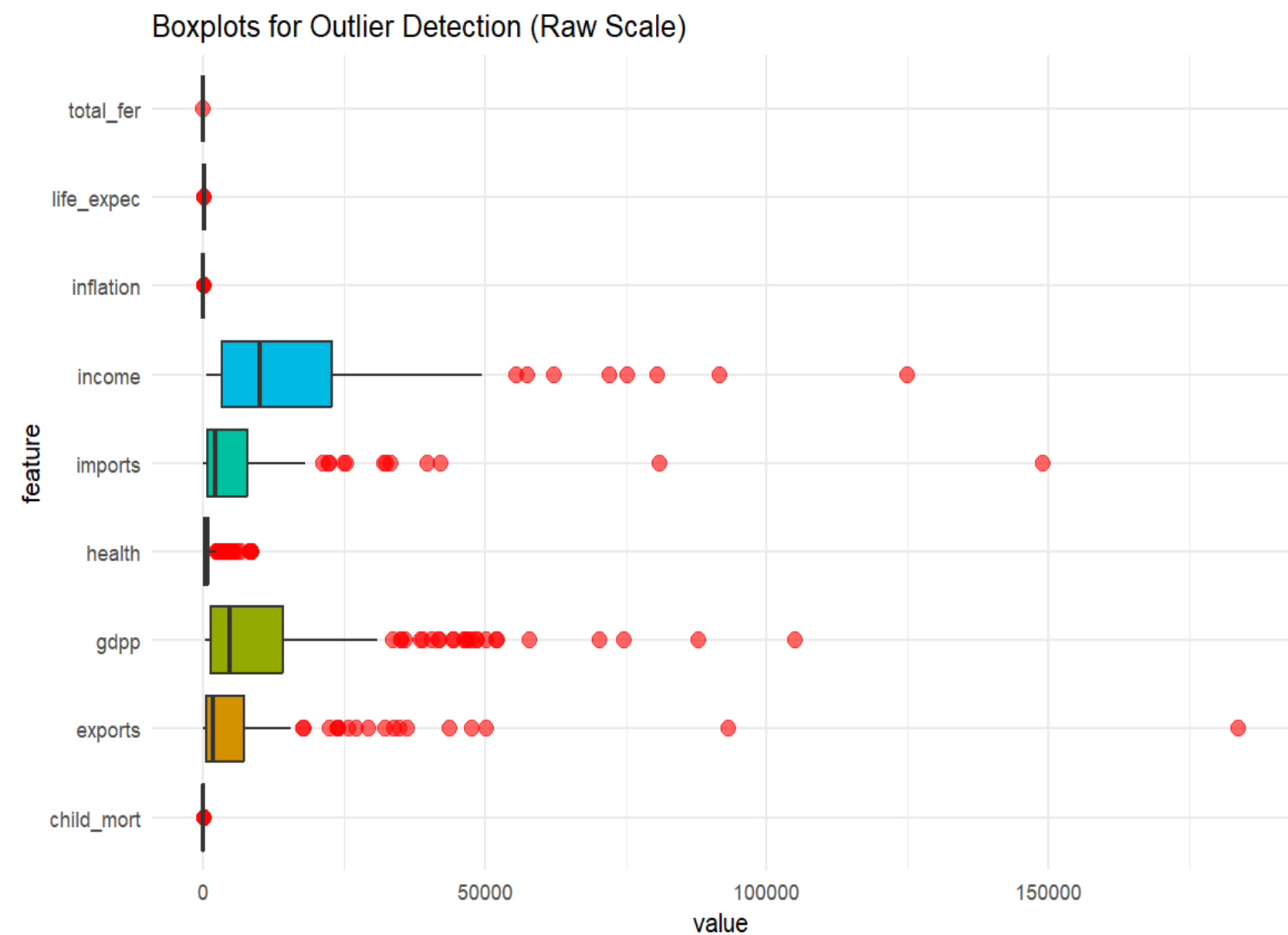
- ❑ **Univariate outliers**: inspected each feature individually using boxplots.
- ❑ **Multivariate outliers**: computed an overall outlier score per country using standardized features:

$$\text{outlier score} = \sqrt{\sum_j z_{ij}^2}$$

(Euclidean distance from the "average country" in Z-score space)



- ❑ **Luxembourg** is the most extreme; **Nigeria** is also very extreme but for a different profile.
- ❑ **Luxembourg & Singapore** stand out strongly and later form a **tiny separate cluster (2 countries)** in both hierarchical and K-means results.
- ❑ Other outliers (e.g., **Qatar, Switzerland, Norway, Ireland, Denmark, USA**) are extreme mainly due to **very high development/wealth indicators**.
- ❑ **Nigeria and Haiti** are extreme due to **high-need conditions** (e.g., high child mortality / low development), **showing that “outlier” means unusual — not only rich countries**
- ❑ Shows the **countries farthest from the “average country”** after Z-score standardization (overall unusual profiles).

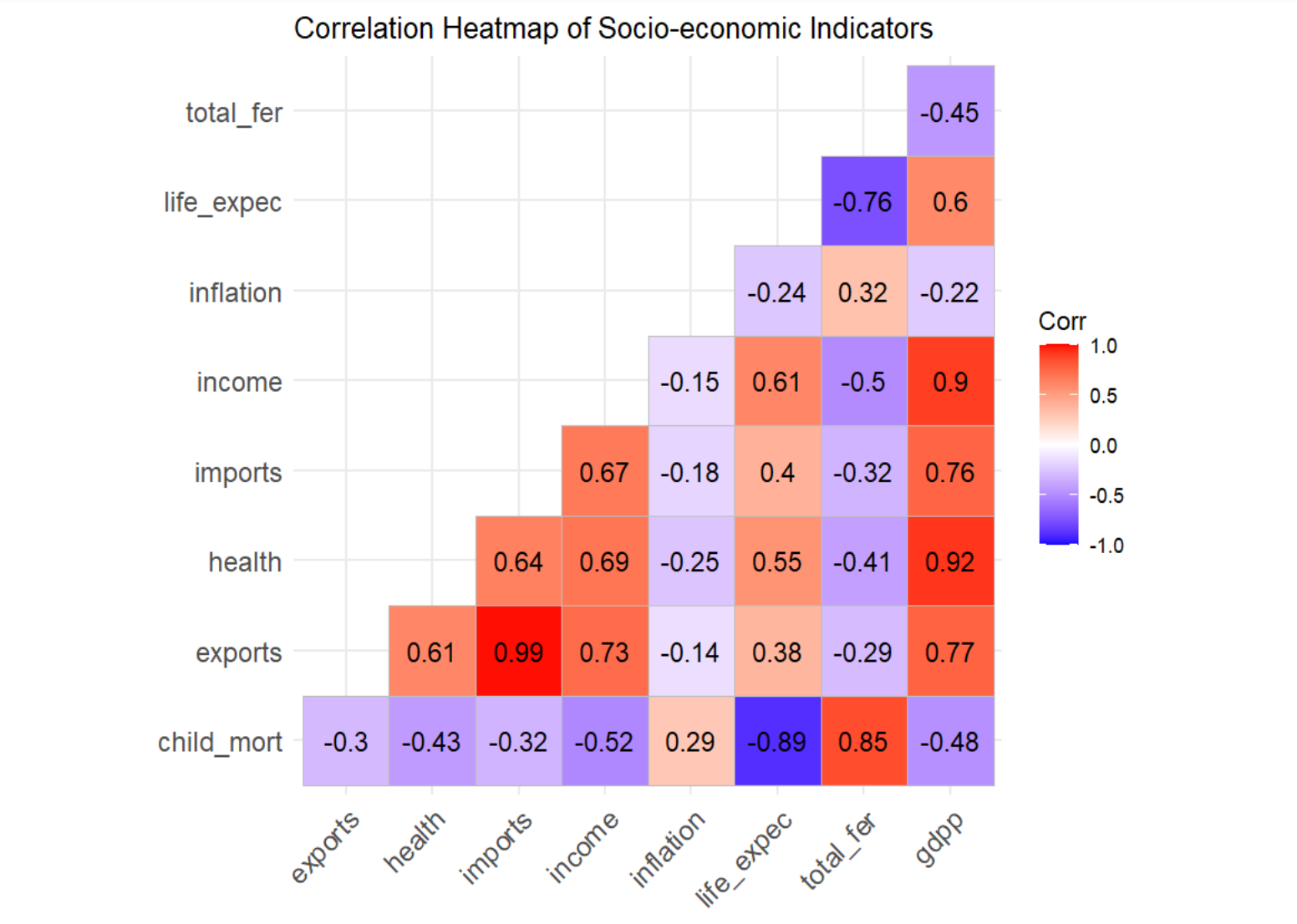


- What the plot shows:
- ❑ Each row is a feature (gdpp, income, exports, imports, health, etc.)
 - ❑ Red points = outliers beyond the whiskers.
- Key insights:**
- ❑ **Income, gdpp, exports, imports** have **many high-end outliers** → consistent with global inequality (a few very high-income/trade-heavy countries).
 - ❑ **health (absolute)** also shows outliers because you converted %GDP into absolute values; rich countries naturally spend much more in absolute terms.
 - ❑ Variables like **life_expec** and **total_fer** show *less extreme scale* compared to economic variables, but still contribute to separating development patterns.

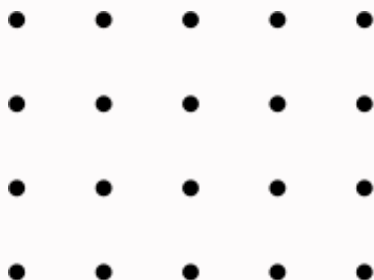
Correlation Structure

Correlation Heatmap: Relationships Among Socio-economic & Health Indicators:

- ❑ **Trade aligns with wealth:**
exports and imports correlate (~ 0.73) and both rise with gdp ($\sim 0.76\text{--}0.77$)
→ stronger economies trade more.
- ❑ **Vulnerability signals are linked:**
child_mort is strongly negative with life_expec (~ -0.89)
and strongly positive with total_fer (~ 0.85);
life_expec and total_fer are negative (~ -0.76)
→ poorer health conditions associate with higher fertility and lower life expectancy.
- ❑ **Wealth indicators move together:**
gdp is strongly correlated with income (~ 0.90)
health spending (~ 0.92) → richer countries earn more and spend more on health.
- ❑ **Inflation is less connected:** correlations are weaker/mixed → inflation reflects macro instability and doesn't fully follow the wealth–health gradient.



Key takeaway: Strong correlations $\Rightarrow$ redundancy $\Rightarrow$ PCA is an appropriate, interpretable dimensionality reduction choice before hierarchical clustering.



PCA Results: How many components to keep (Dimensionality Reduction for clustering)

Using **PCA** as a **dimensionality reduction (DR)** method to compress the 9 standardized variables into a smaller set of components **while keeping most of the information (variance)**.

This also makes **distance-based clustering (Hierarchical/Ward)** more stable and less noisy

PCA for Dimensionality Reduction (DR): Component Selection

- ❑ Data were **standardized (Z-score)** before PCA so all variables contribute fairly.
- ❑ PCA produces orthogonal components ranked by explained variance.
- ❑ We choose **k = 4 PCs** because this is the first point where **cumulative variance $\geq 90\%$** .

Variance results (from the output):

PC1 = **58.94%**

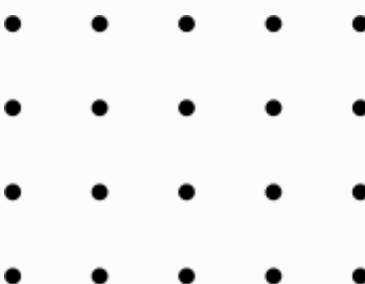
PC2 = **18.45%** → cumulative **77.38%**

PC3 = **9.91%** → cumulative **87.29%**

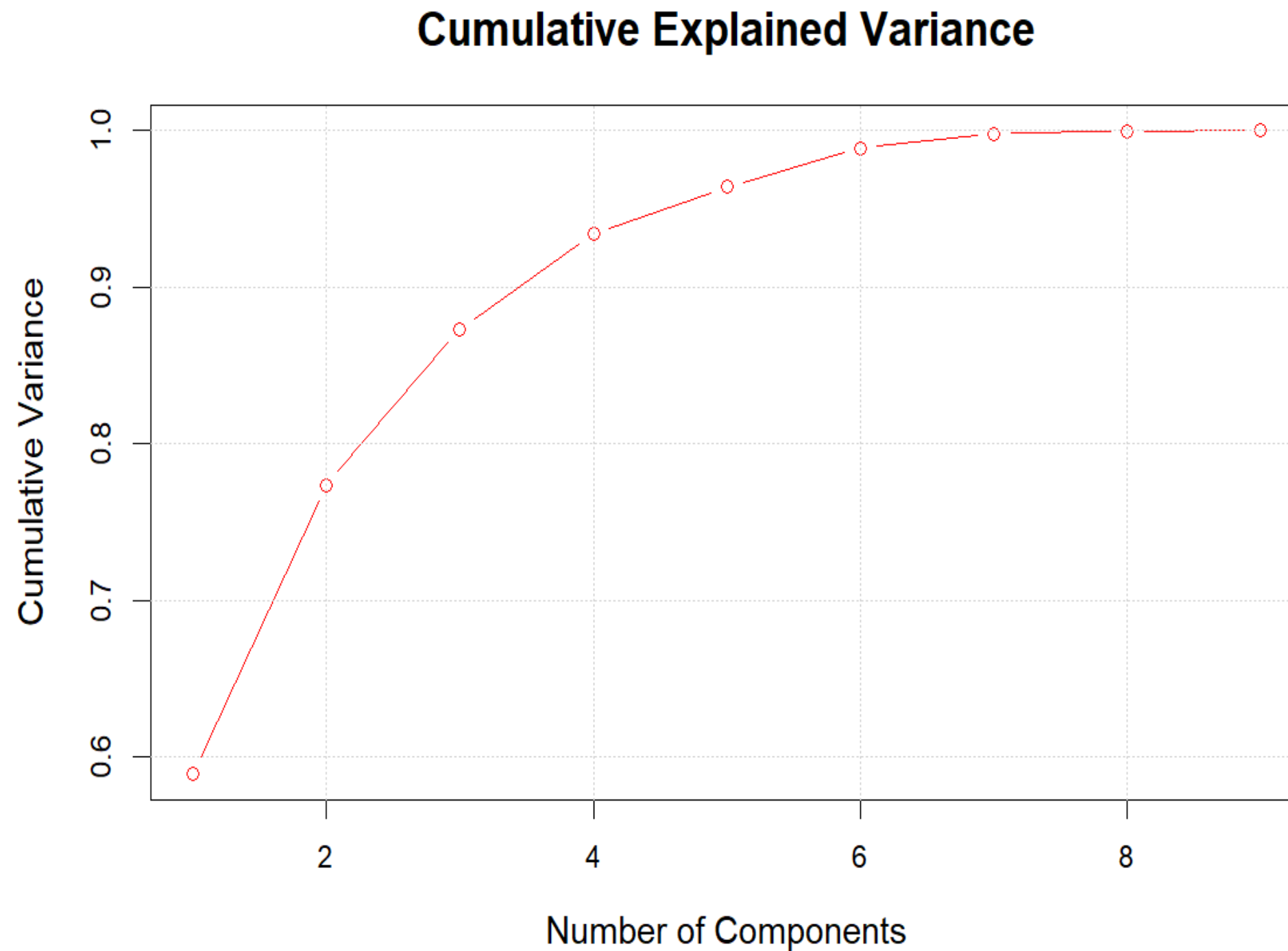
PC4 = **6.07%** → cumulative **93.37%** 

Therefore: **retain 4 PCs** ($k = 4$)

- This PCA step is the **Ddimension reduction baseline** against which the **autoencoder DR** will be compared.
- The reduced PCA representation (**4 dimensions**) becomes the input for **hierarchical clustering**, keeping clustering based on **meaningful structure**, not raw-scale dominance.



Cumulative Explained Variance plot



- ❑ The curve rises sharply at the beginning → meaning **most structure in the data is captured by the first few PCs**.
- ❑ The “knee” happens early: by **PC2** we already capture **~77%**, and by **PC4** we cross **90%**.
- ❑ After PC4, the curve becomes almost flat → additional components add **very small marginal information**, but add complexity and noise.

Autoencoder for Dimensionality Reduction (Non-linear DR)

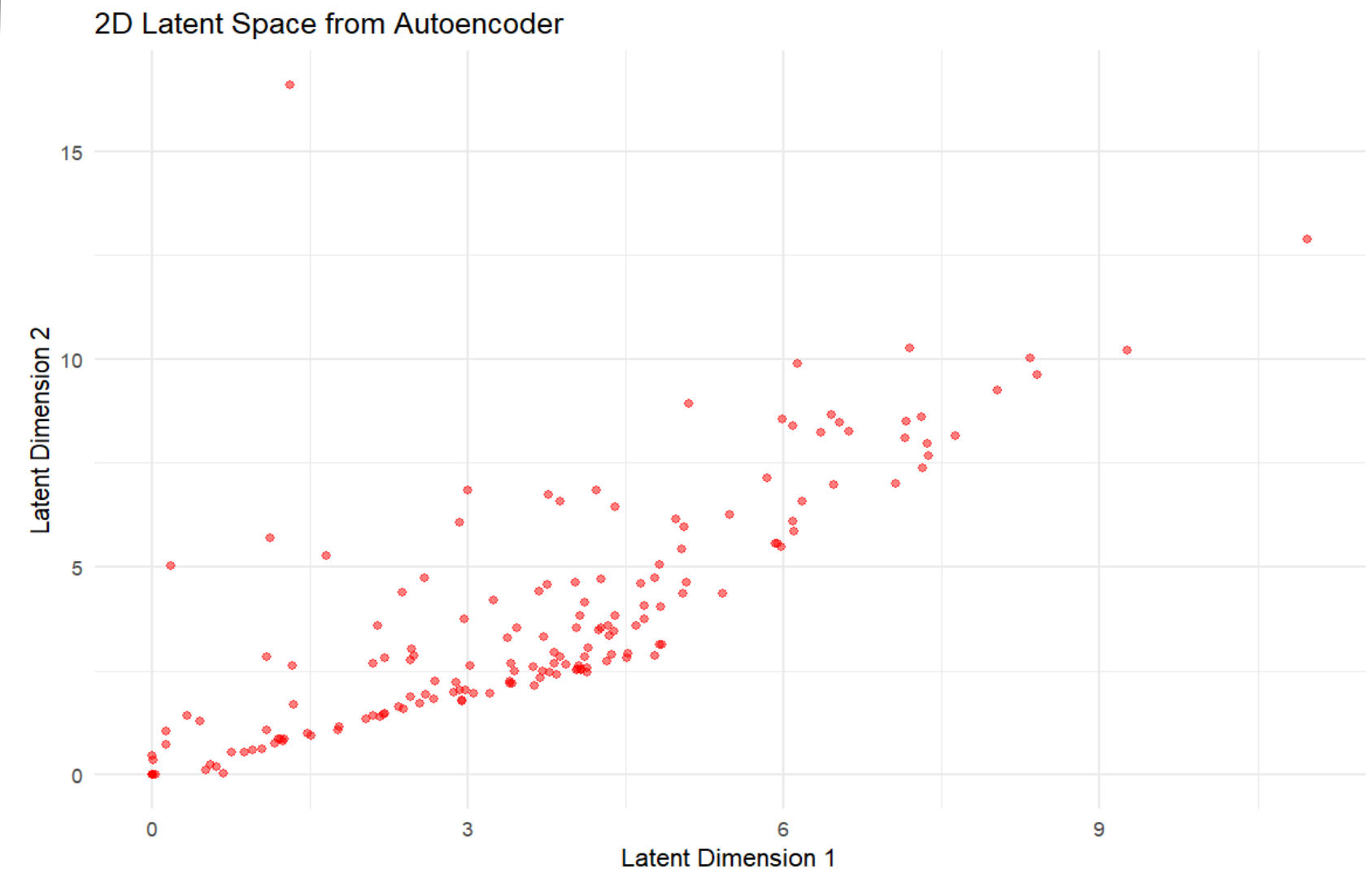
=> Autoencoder-based Dimensionality Reduction (latent_dim = 4)
*Showing the **second Dimension Reduction approach** (Autoencoder) and what it produced, so we can fairly compare it with PCA before clustering.

Model idea:

- ❑ An **autoencoder** learns a compact **latent representation** by reconstructing the original standardized inputs.
- ❑ It can capture **non-linear structure** (unlike PCA which is linear).

Architecture & training:

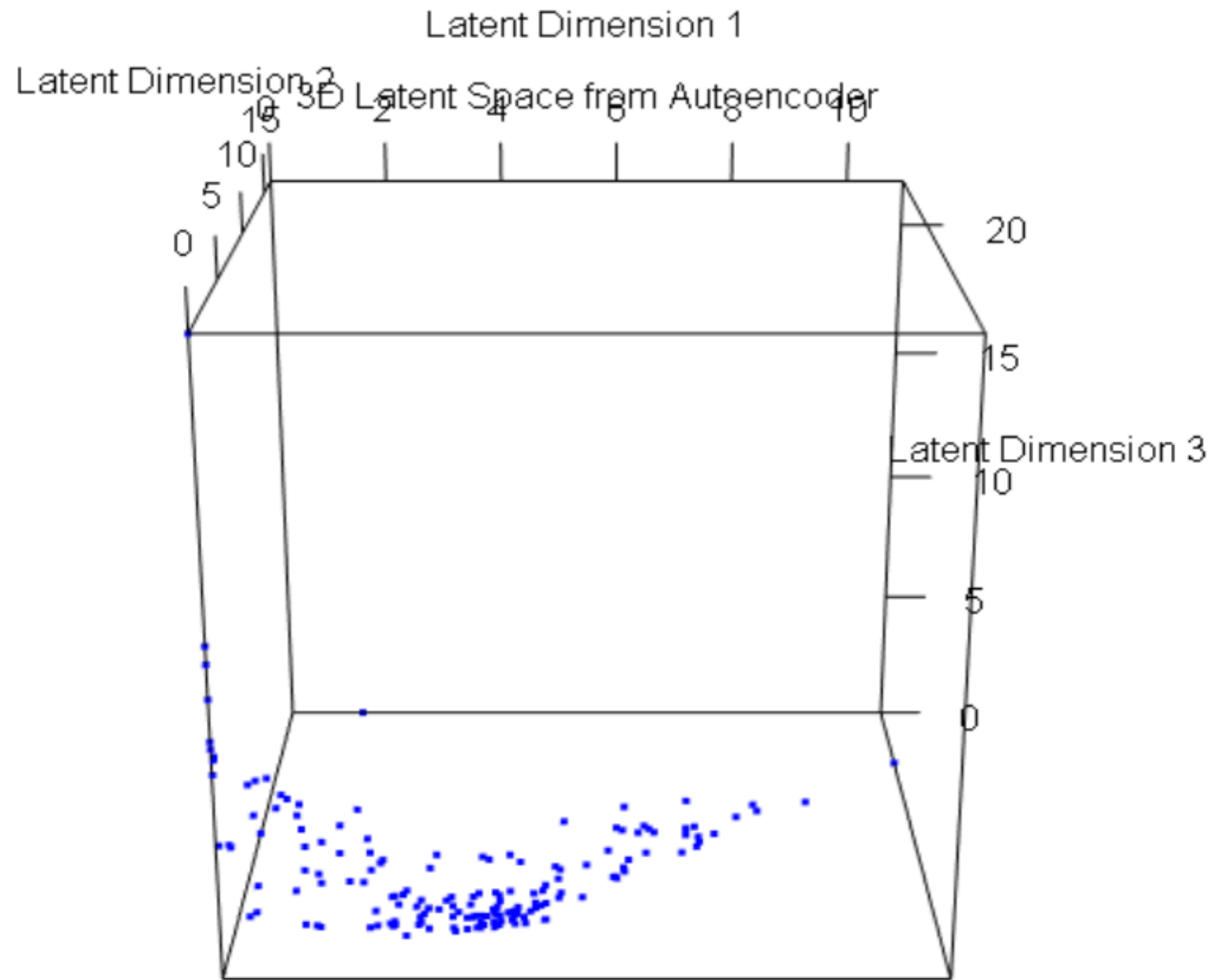
- ❑ Input: **9 standardized features (Z-scores)**
- ❑ Encoder: Dense(128, ReLU) → **Latent space: 4 dims (ReLU)**
- ❑ Decoder: Dense(128, ReLU) → Output: 9 dims (Sigmoid)
- ❑ Loss: **MSE**, Optimizer: **Adam**
- ❑ Training: **50 epochs**, batch size **32**, validation split **0.20**
- ❑ Training loss decreased from **~1.26** to **~0.615**; validation loss to **~0.544** → indicates stable learning (no obvious divergence)



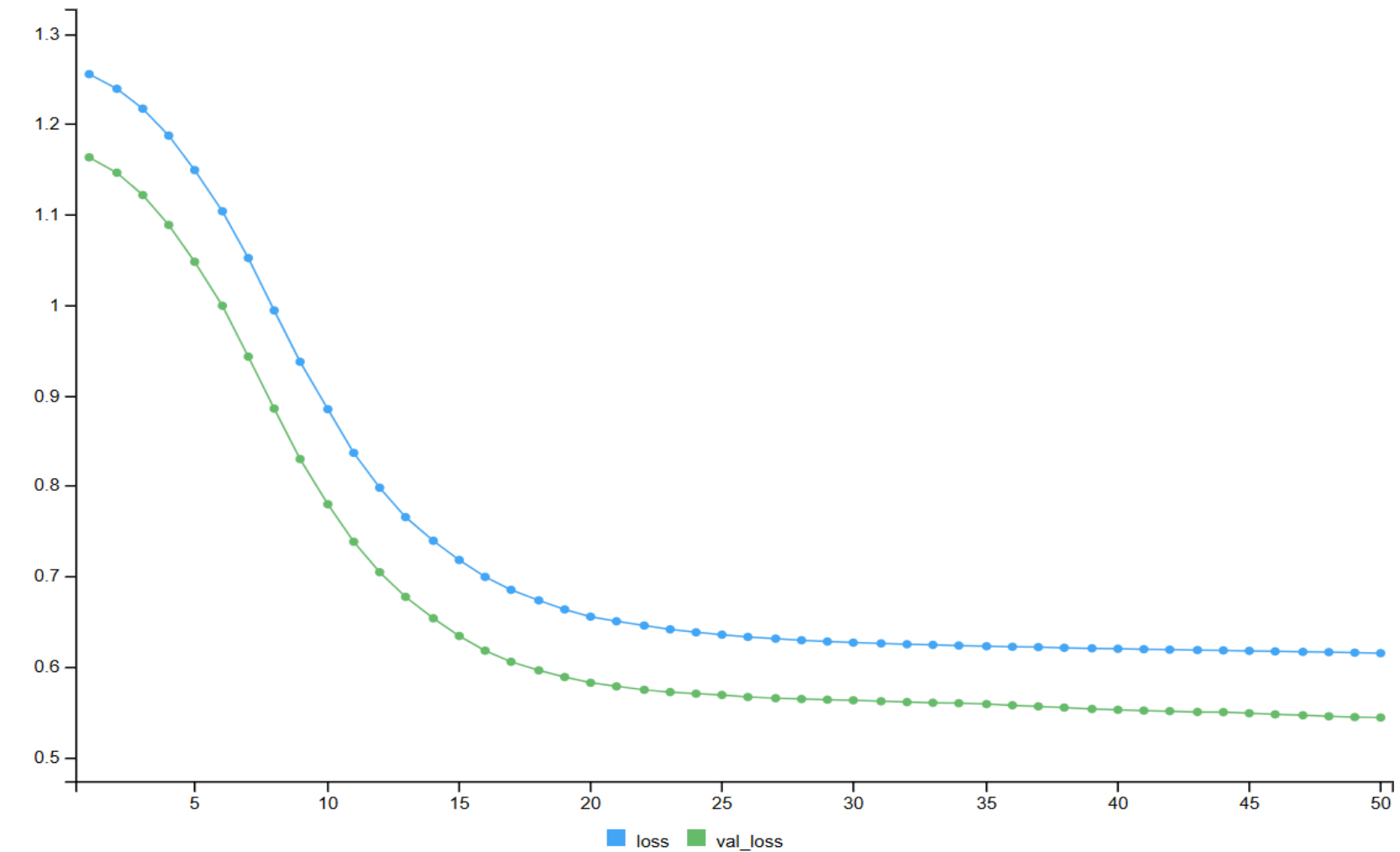
- ❑ Each dot is a **country**, projected into the **autoencoder's latent space**.
- ❑ The axes are **learned latent dimensions**, not directly interpretable like PCs.
- ❑ The points lie along a **smooth, almost monotonic "manifold/curve"** rather than forming clearly separated blobs.
- ❑ If the dataset had strong non-linear structure, we would expect the autoencoder latent space to show **clearer separation** than PCA.

Key Takeaway: *The autoencoder learns a compact 4D representation and captures the main data gradient, but it does not reveal substantially stronger non-linear separation than PCA in this dataset.*

Autoencoder 3D latent space



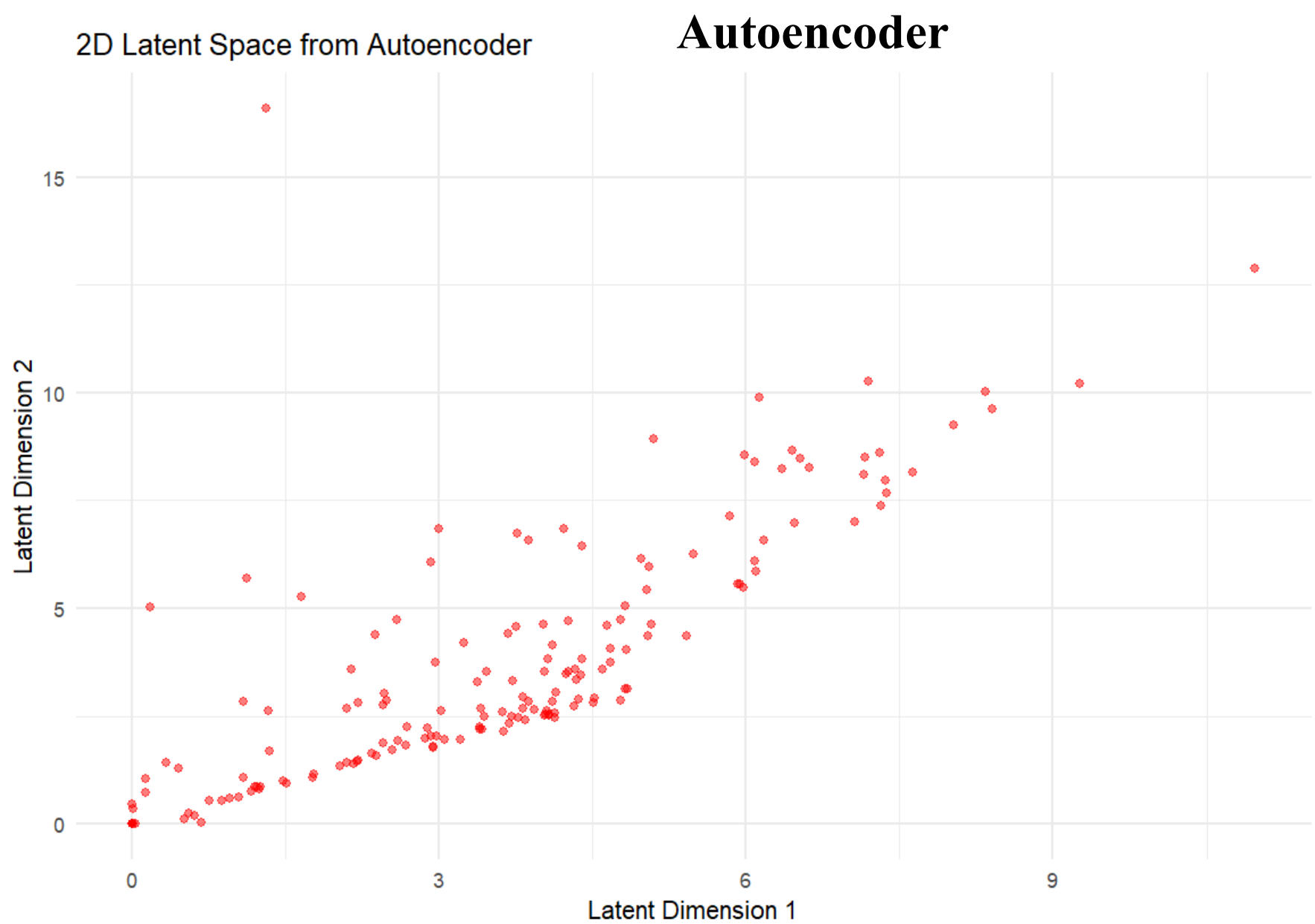
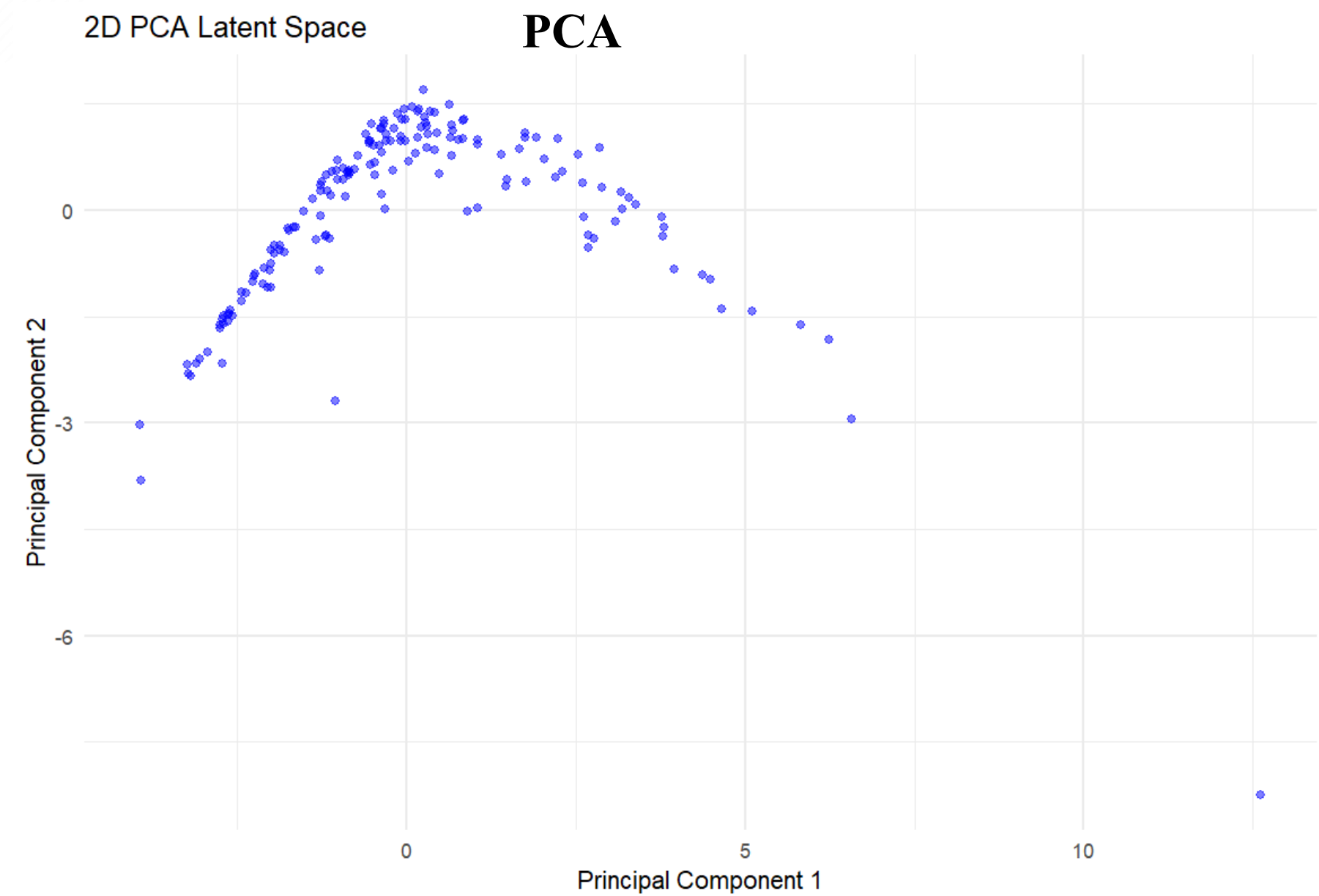
Learning Curve: Autoencoder Loss (Train vs Val)



- ❑ **Training and validation loss drop sharply at the start**, showing the autoencoder quickly learns a useful compressed representation.
- ❑ **After the early epochs, both curves plateau**, meaning learning stabilizes and later improvements are smaller.
- ❑ **Validation loss stays close to training loss (no large gap)**, suggesting no strong overfitting and a reliable latent space for clustering/visualization.
- ❑ **Final validation loss ≈ 0.54** , indicating stable reconstruction error.

PCA vs Autoencoder: Latent Space Comparison (2D)

Comparing Dimensionality Reduction (DR): PCA vs Autoencoder (2D Latent Spaces):



PCA vs Auto encoder Continuing...

PCA: linear DR → components maximize explained variance.

Autoencoder: non-linear DR → latent space learned by minimizing reconstruction error.

What to look for in the plots:

- ❑ In both projections from the previous slide, countries form a **continuous structure/gradient** rather than perfectly separated clusters.
- ❑ This suggests the dominant pattern is a **development continuum** (wealth/health vs vulnerability), consistent with the correlation heatmap.
- ❑ The autoencoder does not create a clearly more separated structure than PCA in 2D → indicates **non-linearity is not dominant** in this dataset.

Decision statement (based on the results):

- ❑ **Chosen Dimension Reduction for clustering: PCA (k = 4 PCs)**
- ❑ **Reason:** strong linear correlations + PCA reaches **~93% variance with 4 PCs**, giving an interpretable and stable representation for hierarchical clustering.

Key Takeaway:

Both methods capture a similar development gradient; PCA is preferred here for interpretability and strong linear structure.

Hierarchical Clustering Setup (Ward's Method on PCA)

Hierarchical Clustering (Ward.D2) on PCA-Reduced Data

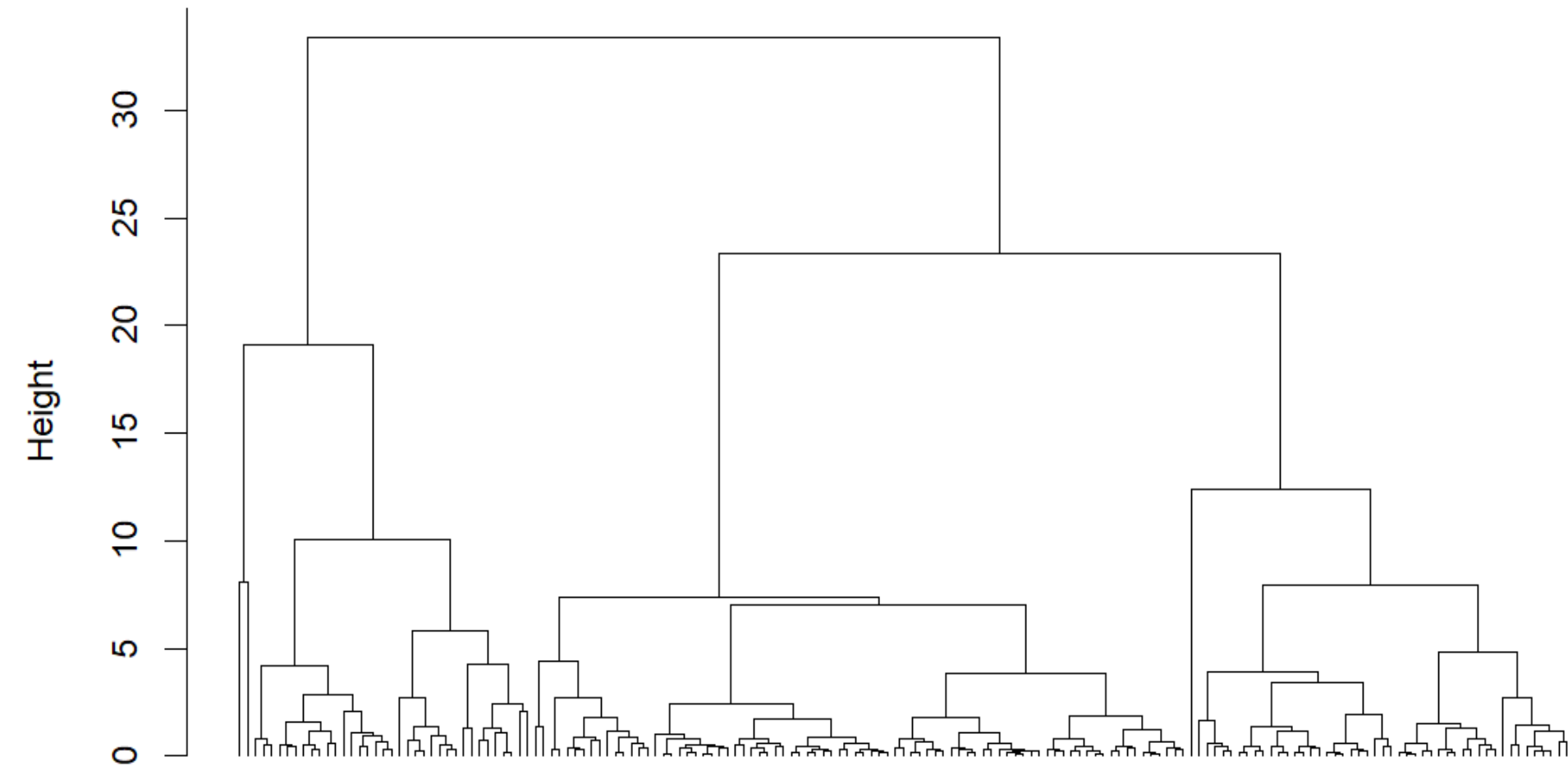
- ❑ **Clustering type: Agglomerative hierarchical clustering** (starts with each country alone, then merges step-by-step).
- ❑ **Input data: PCA scores (4 PCs, ~93% variance retained)** to reduce noise and redundancy before clustering.
- ❑ **Distance measure: Euclidean distance** computed in PCA space.
- ❑ **Linkage method: Ward.D2**, which merges clusters by minimizing the increase in **within-cluster variance** (compact clusters).
- ❑ **Output: a dendrogram**, where merge **height** shows how dissimilar clusters are when they combine.

Standardize → PCA (4 PCs) → Distance → Ward Hierarchical Clustering → Dendrogram

Key Takeaway: Ward hierarchical clustering on PCA scores groups countries by overall similarity while controlling within-cluster variance.

Dendrogram (Ward's Method)

Hierarchical Clustering Dendrogram (Ward's Method)



hclust (*, "ward.D2")

- ❑ Each leaf at the bottom is a **country** (labels are hidden for readability).
- ❑ The **height** where branches merge = **dissimilarity** (higher merge height → more different groups).
- ❑ Big vertical jumps suggest **natural separation points** → these guide the choice of **number of clusters (k)**.

Choosing $k = 4$ (cutting the dendrogram)

How k is selected

- ❑ The dendrogram shows **large jumps in merge height**, which indicate natural separation between groups.
- ❑ Cutting the tree at the level that preserves those jumps gives **well-separated clusters**.

Chosen number of clusters

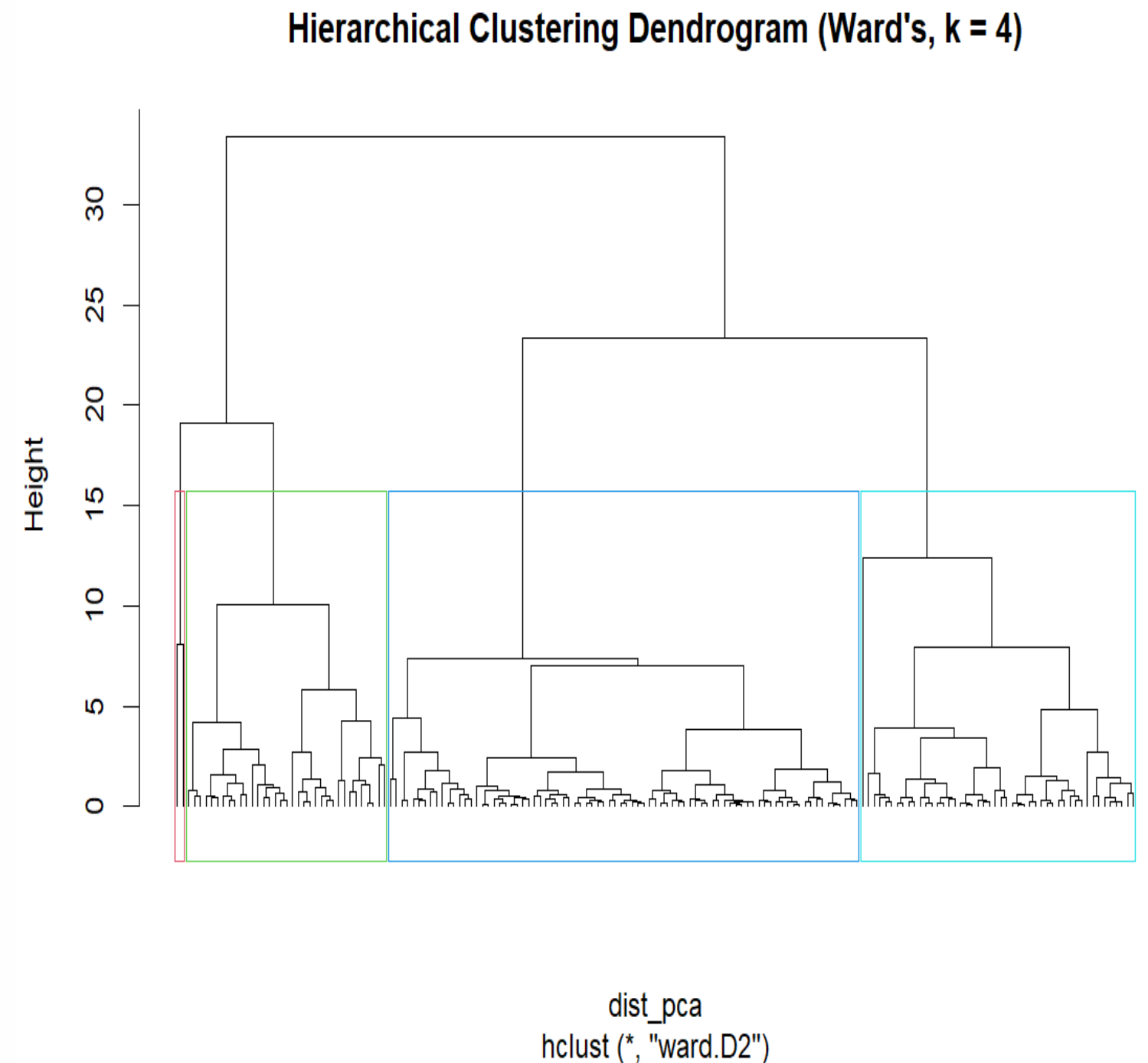
- ✅ **Final choice: $k = 4$**

What happens after cutting

- ❑ Each country receives a cluster label (1–4).
- ❑ Cluster sizes (from your output):
- ❑ **$C1 = 48$, $C2 = 82$, $C3 = 35$, $C4 = 2$**

Interpretation from the dendrogram)

- ❑ At $k = 4$, the dendrogram produces:
- ❑ one **very small cluster (2 countries)** → extreme high-development outliers (Luxembourg & Singapore)
- ❑ other clusters representing low-, middle-, and high-development groups.
- ❑ The small cluster is meaningful (not an error): it reflects **exceptional economic profiles**.



Key Takeaway:

$k = 4$ is selected from the dendrogram structure and yields interpretable clusters, including a distinct outlier-rich group

Cluster Validation: Silhouette Analysis (Hierarchical, k = 4)

Purpose: Silhouette evaluates how well each country fits its assigned cluster (**cohesion vs separation**).

Interpretation rule:

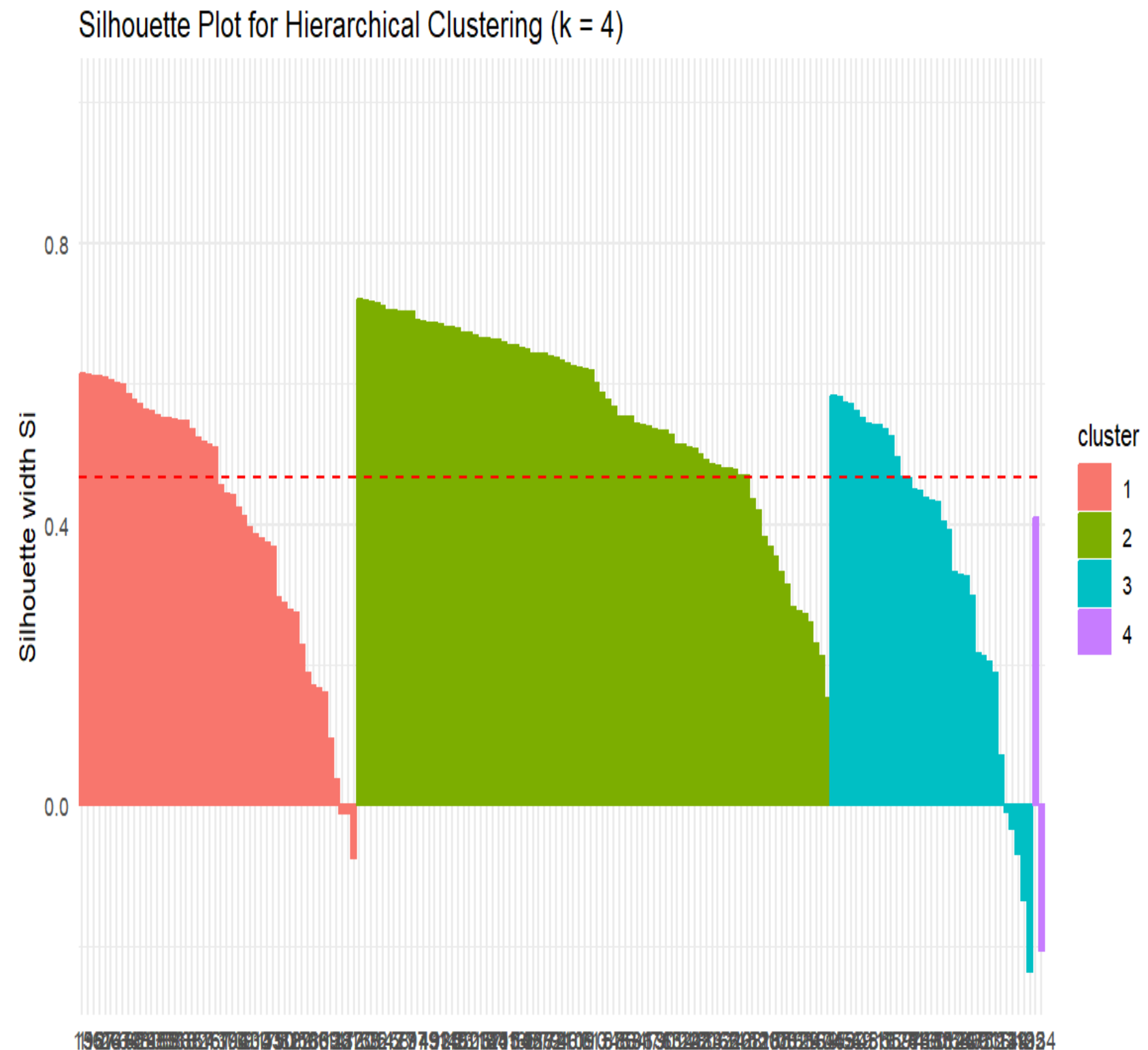
- ❑ **Si close to 1** = well-clustered
- ❑ **Si near 0** = overlap between clusters
- ❑ **Negative Si** = likely misclassified / closer to another cluster

Overall result: Average silhouette ≈ 0.47 $\rightarrow$ *moderate-to-good clustering quality* for k = 4.

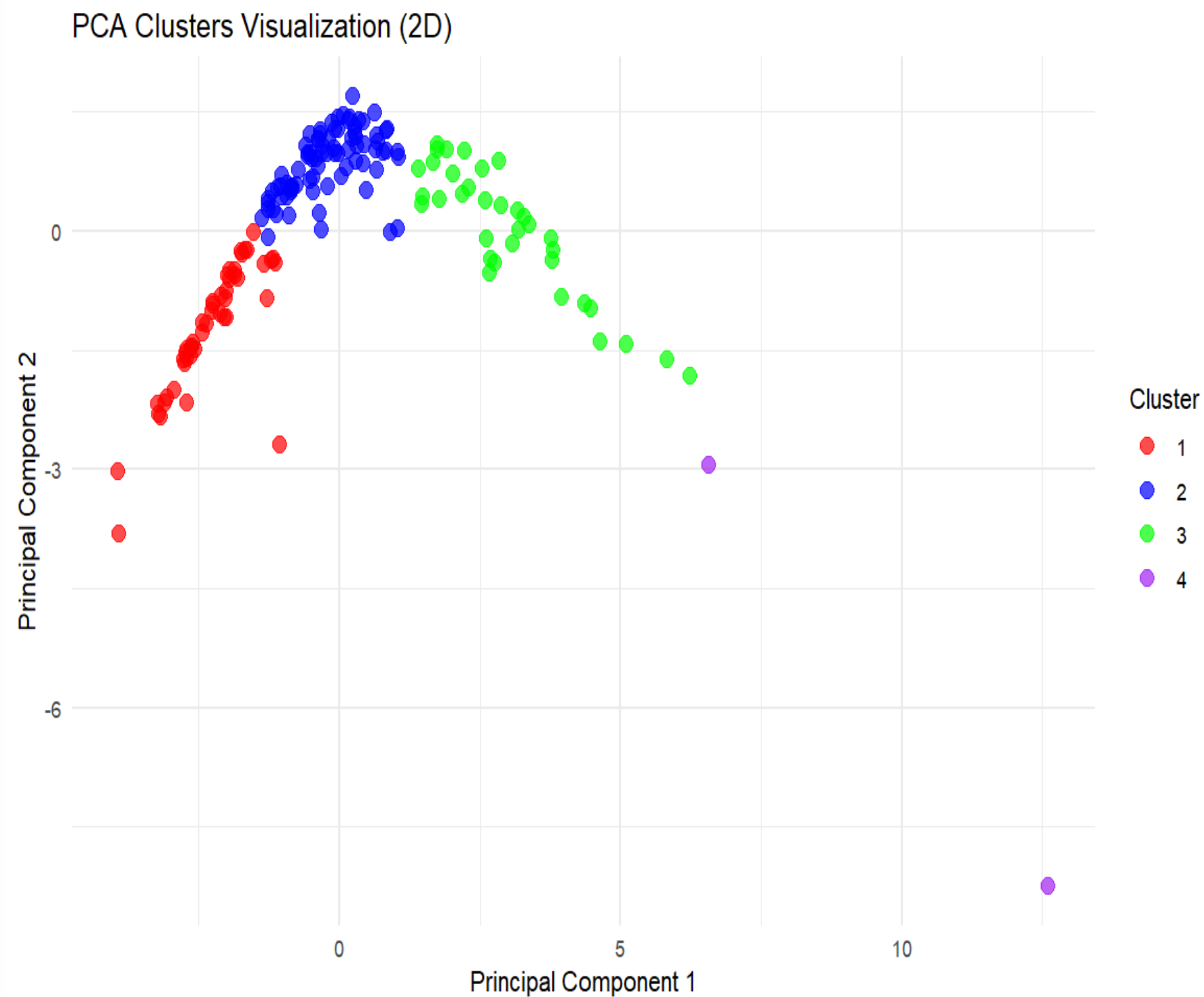
Cluster-level insight (from the output):

- ❑ Cluster 2 performs best (**avg ≈ 0.56**) $\rightarrow$ strong separation
- ❑ Cluster 1 moderate (**avg ≈ 0.41**)
- ❑ Cluster 3 weaker (**avg ≈ 0.35**) $\rightarrow$ more overlap
- ❑ Cluster 4 very low (**avg ≈ 0.10** , size = 2) $\rightarrow$ tiny outlier cluster (Luxembourg & Singapore)

Key Takeaway: k = 4 yields **three meaningful clusters** plus **one small outlier cluster** (Luxembourg & Singapore), and the main clusters show good separation overall.



Visualizing the Hierarchical Clusters in PCA Space (2D)



- ❑ This plot projects countries onto the first two principal components (**PC1 vs PC2**).
 - ❑ Points are colored by the **hierarchical cluster labels (Ward.D2, k = 4)**.
 - ❑ Countries that appear close together are **similar across the socio-economic and health indicators** (in the PCA-reduced space).
 - ❑ The separation you see indicates that hierarchical clustering successfully captures **distinct country profiles**.
- *The PCA map shows a clear development–vulnerability gradient, and the hierarchical clusters align with this structure, including a distinct 2-country outlier cluster.*

Cluster Profiling (Hierarchical Clustering)

Translating the clusters into **real-world meaning** using the original indicators (gdpp, income, health, child_mort, exports).

Cluster 1 — Highest vulnerability / lowest development

- ❑ Lowest **gdpp** and **income**, very low **health spending**
- ❑ Highest **child mortality** (~91.6)
- ❑ These countries are in severe socio-economic and health stress → **top priority for aid**

Cluster 2 — Lower-middle development

- ❑ Moderate **gdpp/income** and **health spending**
- ❑ Child mortality drops strongly (~22) compared to Cluster 1
- ❑ These are emerging/transition countries → targeted support (health systems, stability, services)

Cluster 3 — High development

- ❑ High **gdpp/income**, high **trade** and **health spending**
- ❑ Very low child mortality (~5.4)
- ❑ Strong capacity countries → low priority for humanitarian aid

Cluster 4 — Extreme high-development outliers (2 countries)

- ❑ Extremely high values across **gdpp**, **income**, **exports**, **health**
- ❑ Lowest child mortality (2.8)
- ❑ **Luxembourg & Singapore** form a distinct outlier cluster; exclude from “aid need” ranking (not in need)

Takeaway: Clusters align with a clear development gradient; **Cluster 1** shows the strongest need profile and becomes **the main target for aid prioritization**.

Summary Table (Cluster Mean Profile (Hierarchical))						
Country	Hierarchical Cluster	Avg GDPP	Avg Income	Avg Exports	Avg Health	Avg Child Mortality
	C1	1,909	3,897	879	115	91.60
	C2	6,287	12,305	2,789	393	22.00
	C3	40,177	42,951	19,754	3,678	5.38
	C4	75,800	81,900	138,475	5,002	2.80

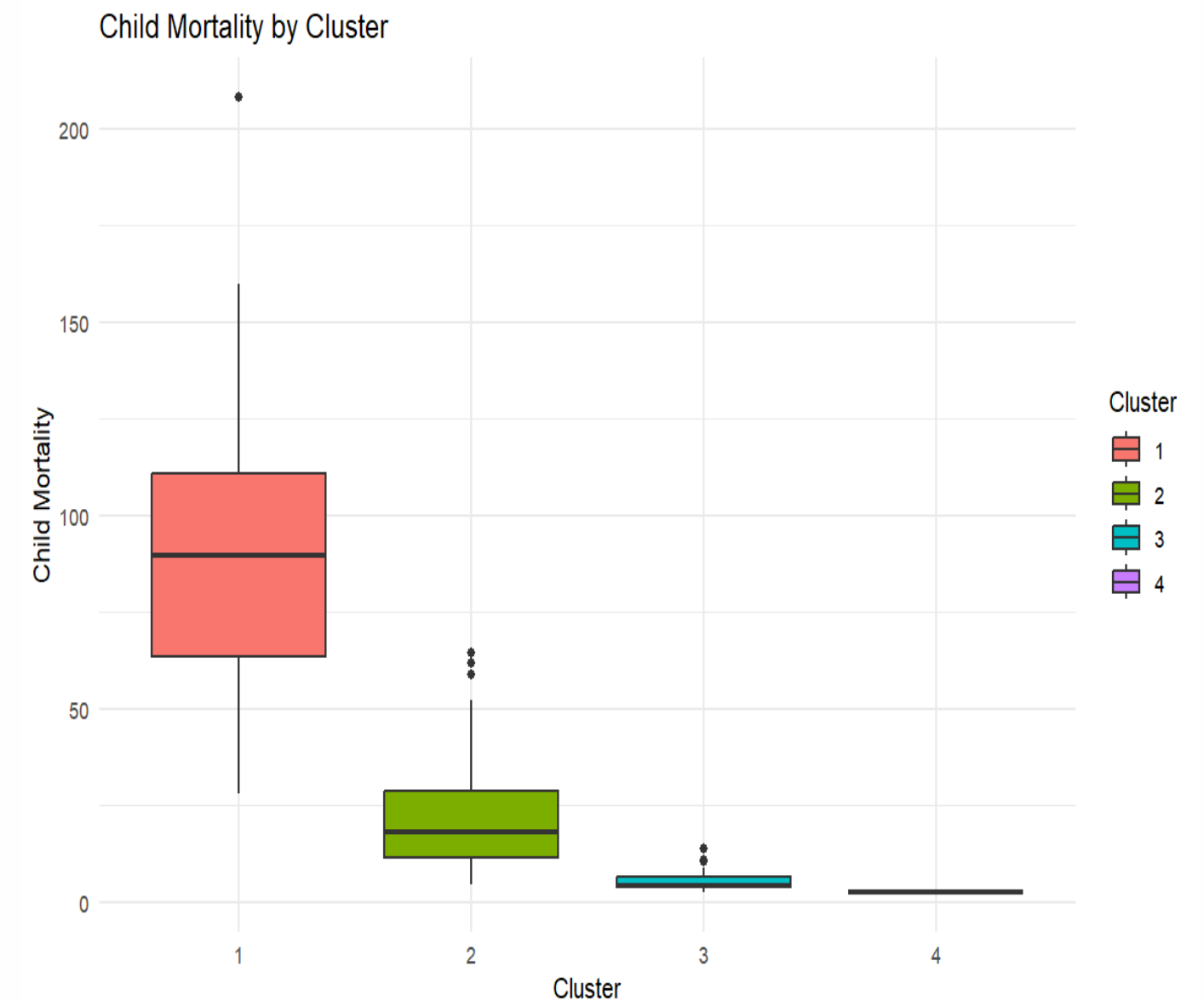
Aid Prioritization: Comparing Key Indicators Across Clusters

Showing why one cluster should be prioritized for aid using clear visuals:

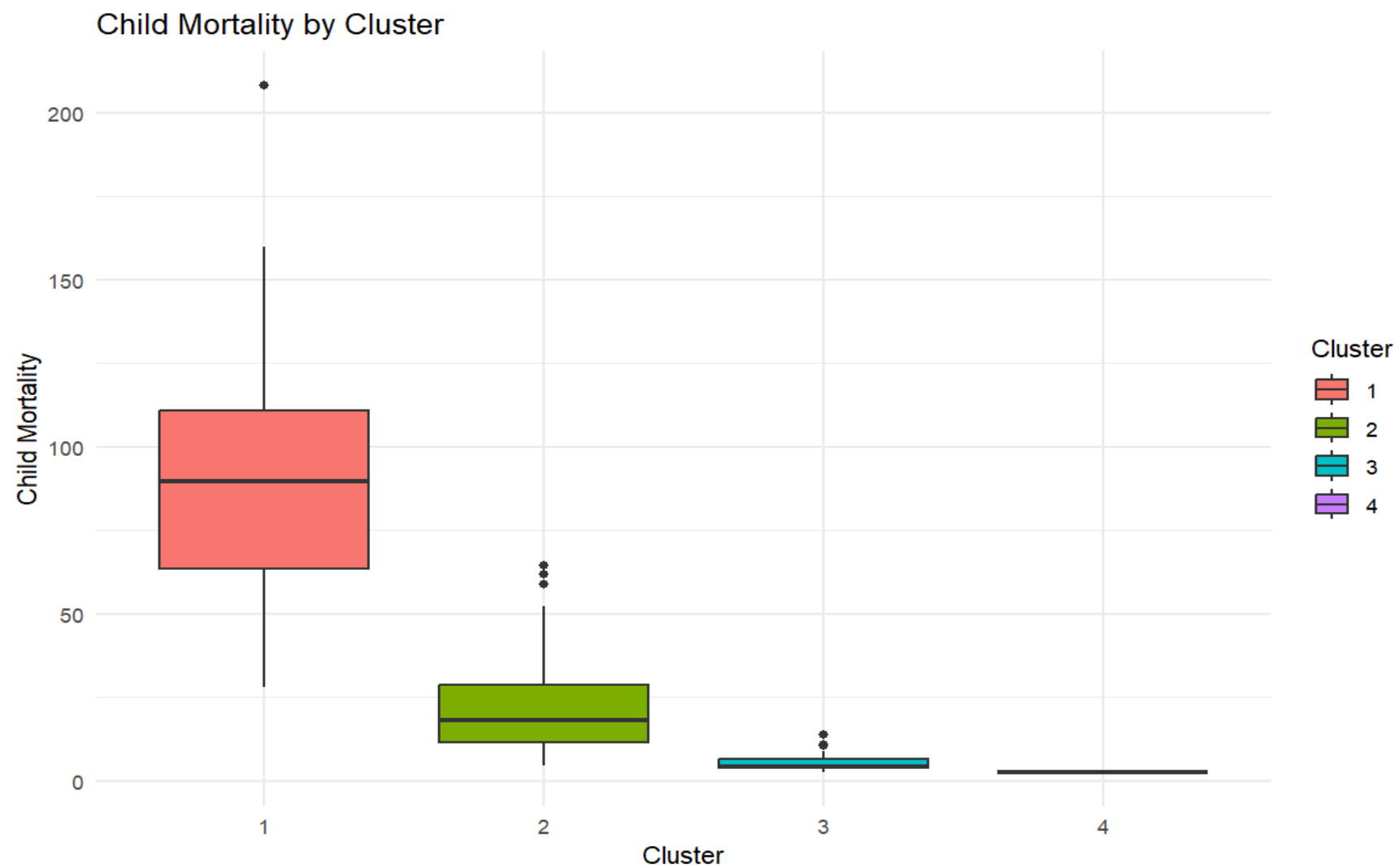
child mortality, GDP per capita, and health expenditure across clusters.

- ❑ **GDP per capita:** Cluster 1 has the lowest values → weakest economic capacity.
- ❑ **Child mortality:** Cluster 1 has the highest distribution → strongest vulnerability signal.
- ❑ **Health expenditure:** Cluster 1 has the lowest spending → limited health-system resources.

Together, these three indicators consistently point to **Cluster 1 as the highest-need group**.



Continuing....



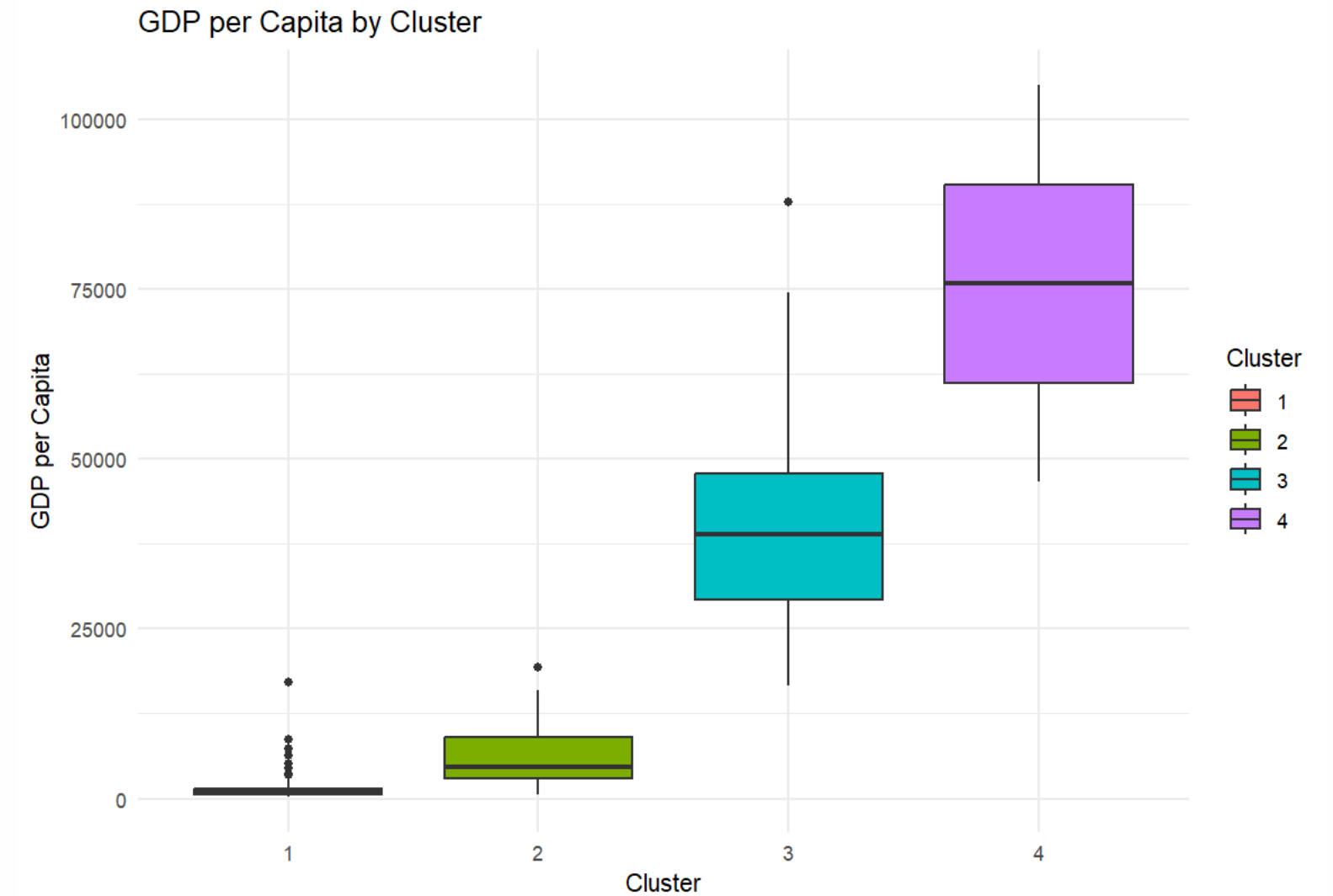
Decision box

✓ **Cluster in most need of aid (Hierarchical): Cluster 1**

❑ Avg child mortality $\approx$ **91.6**

❑ Avg gdpp $\approx$ **1,909**

❑ Avg health $\approx$ **115**



Key Takeaway:

Across all three indicators, **Cluster 1** combines the highest mortality with the lowest economic and health capacity → **priority cluster for HELP International aid planning.**

Aid Prioritization (Who should HELP International fund first?)

Aid Prioritization Using Cluster Evidence (Hierarchical Clusters)

- ❑ Interpretation across the three indicators
- ❑ **Cluster 1 = highest vulnerability:** *very high child mortality + lowest GDP per capita + lowest health spending.*
- ❑ **Cluster 2 = moderate need:** child mortality drops a lot, GDP/health improve, but still clearly below developed groups.
- ❑ **Cluster 3 = developed:** strong GDP and health spending, low child mortality → **lower aid priority.**
- ❑ **Cluster 4 = extreme wealthy outliers** (Luxembourg & Singapore): very high GDP/exports/health, lowest child mortality → **not aid targets.**

Decision box

- ✅ **Priority cluster for aid: Cluster 1 (Hierarchical)**
- ❑ **Reason:** It concentrates countries with **low economic capacity + weak health investment + high child mortality**, matching HELP International's mission and the “most in need” definition.

Country-Level Prioritization (Within the High-Need Cluster)

Ranking Countries by “Need Score” for Targeted Aid Allocation

Goal: After identifying **Cluster 1** as highest-need, now showing **which specific countries** inside it should be **funded first**.

A composite “Need Score” was computed using **three key indicators**:

- ❑ **Higher child mortality** → increases need
- ❑ **Lower GDP per capita** → increases need
- ❑ **Lower health expenditure** → increases need

All three indicators were **standardized (z-scores)** to make them comparable.

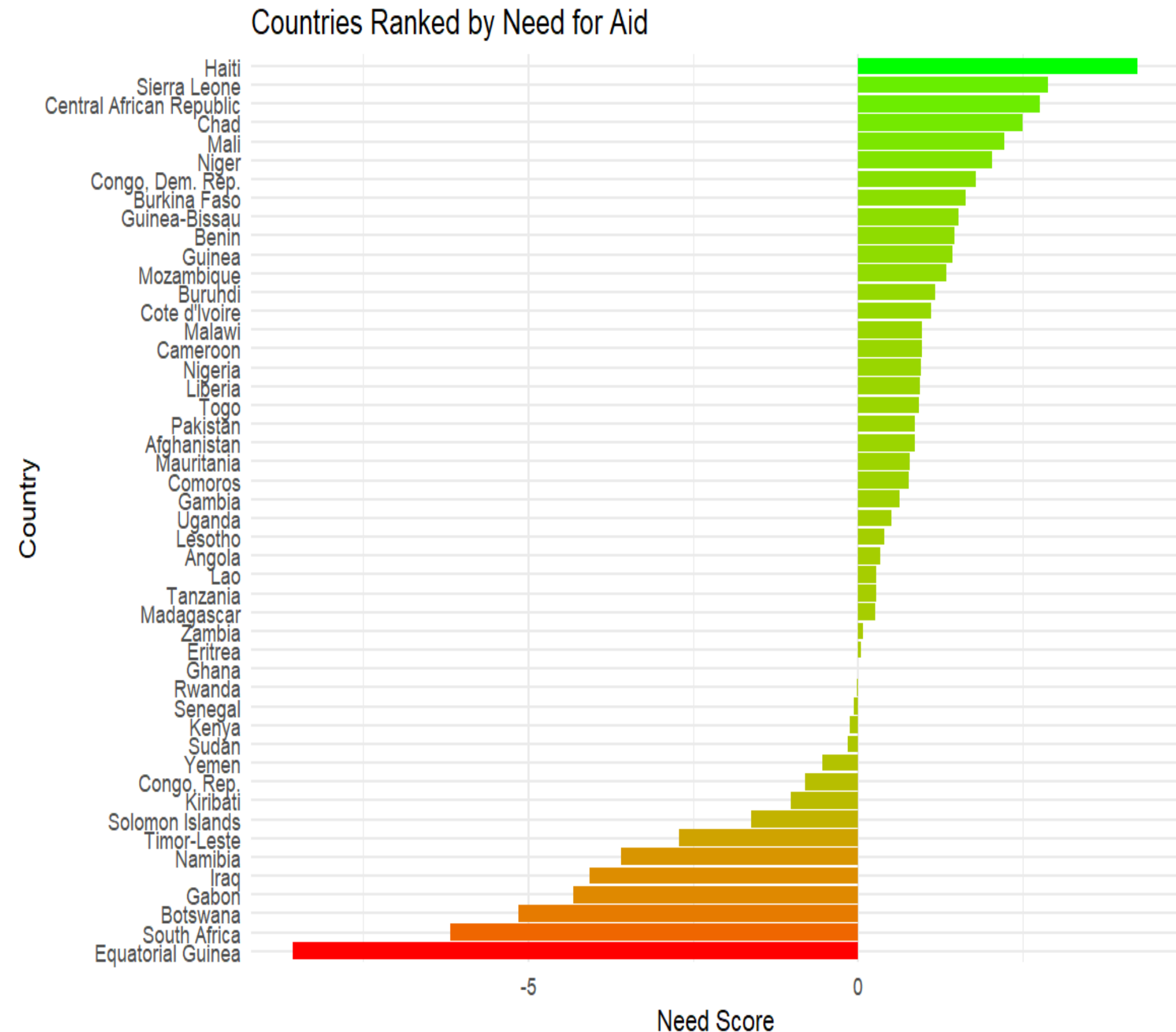
- ❑ **Final score logic (direction):**

Need Score = (+ child_mort) – (gdpp) – (health)

✓ Higher score = **higher urgency for aid**

- ❑ Countries on top combine **very high mortality** with **very low economic and health capacity** eg. **Haiti (highest), Sierra Leone, Central African Republic, Chad and Mali**
- ❑ This ranking supports **targeted allocation** rather than spreading money evenly across the whole cluster.

Key Takeaway: Cluster-level prioritization identifies the “**most in need**” group, while the **Need Score ranks** countries inside that group for practical funding decisions



Final Recommendations for HELP International (\$10M Allocation Plan)

1) Cluster-based priority

✓ **Cluster 1** (Hierarchical) = Highest Priority

- ❑ Lowest economic capacity (gdpp, income)
- ❑ Lowest **health expenditure**
- ❑ Highest **child mortality** → strongest vulnerability signal

● **Cluster 2** = Medium Priority

- ❑ Moderate gdpp/income and health spending
- ❑ Child mortality clearly lower than Cluster 1
- ❑ Needs support for resilience and system strengthening

● **Cluster 3** = Low Priority (high-capacity countries)

- ❑ High gdpp/income and high health spending
- ❑ Very low child mortality
- ❑ Not a direct humanitarian aid target (could be partners/benchmarks instead)

✗ **Cluster 4** = Exclude from aid allocation

(Luxembourg & Singapore)

- ❑ Extreme high-development outliers, not aid recipients

2) Suggested allocation logic (how to use \$10M)

- **70–80%** → **Cluster 1** (life-saving & urgent needs)
 - ❑ maternal/child health services
 - ❑ nutrition + vaccination coverage
 - ❑ basic healthcare access and essential medicines
- **20–30%** → **Cluster 2** (prevention + resilience)
 - ❑ strengthen primary healthcare systems
 - ❑ water/sanitation and community health programs
 - ❑ capacity-building to reduce risk of sliding toward Cluster 1
- **0%** → **Cluster 3 and Cluster 4** (*not aid targets*)

Limitations and Improvements of the Clustering Approach

A) Limitations of the clustering approach

- ❑ **Sensitivity to outliers:** extreme countries (e.g., Luxembourg & Singapore) strongly affect distances and form a tiny cluster, influencing the dendrogram structure.
- ❑ **Choice of k is partly subjective:** k = 4 is supported by dendrogram + silhouette, but other cuts could also be defensible eg. K=3, which will make those outlier merge into the broader high-development cluster rather than forming a separate group.
- ❑ **Cluster quality is moderate (not perfect):** average silhouette for hierarchical clustering is ~0.47, meaning clusters are reasonable but some overlap remains.
- ❑ **Distance-based method depends on scaling & metric:** Euclidean distance after Z-score scaling can still over-weight high-variance patterns.
- ❑ **Limited feature space:** results only reflect the given indicators; omitted factors (governance, inequality, conflict, education quality) could change the grouping and the “aid need” story.

• • • • • • •
• • • • • • •
• • • • • • •

(B) Alternative methods to enhance the analysis

- ❑ **Robustness checks:** compare linkage methods (complete/average) and distance metrics (Manhattan/correlation distance).
 - ❑ **Stability analysis:** bootstrap or subsampling to test whether cluster assignments remain consistent.
 - ❑ **Handle outliers explicitly:** robust scaling, winsorizing, or clustering with/without the extreme outliers to assess impact.
 - ❑ **Better cluster validation:** use additional indices (Calinski–Harabasz, Davies–Bouldin) not only silhouette.
- Alternative clustering methods:**
- ❑ Gaussian Mixture Models (GMM) for probabilistic clusters (soft assignment)
 - ❑ DBSCAN/HDBSCAN if clusters are non-spherical and density-based
 - ❑ Interpretability upgrade: use PCA loadings to label components and explain what each cluster is “high/low” in, beyond means.



THANK YOU